

Trust-Sleep: A Trust-Aware Clinical Decision Framework with Selective Prediction for Sleep Apnea Screening

Abstract—Deep learning has shown promise for automated sleep apnea screening from polysomnography (PSG), but most existing approaches emphasize predictive accuracy while providing limited support for determining whether individual predictions should be trusted in clinical practice. In safety-critical settings, clinicians require not only accurate predictions but also principled information about uncertainty, reliability, and when automated outputs should be deferred for further review. To address this need, we propose Trust-Sleep, a trust-aware clinical decision-support framework for sleep apnea screening that integrates predictive modeling, reliability assessment, and selective prediction within a unified architecture. The framework combines a multi-branch deep learning model with event-centric physiological representations and introduces three complementary trust measures: *Clinical Decision Trust (CDT)*, which aggregates predictive confidence, uncertainty estimation, conformal reliability, physiological evidence, signal quality, and related decision-support factors; *Clinical Explanation Reliability Assessment (CERA)*, which evaluates explanation stability, temporal consistency, physiological alignment, and related automated reliability criteria; and *Clinical Acceptance Score (CAS)*, which supports selective acceptance or clinician escalation decisions. Experiments on a pediatric PSG analysis export containing 240,043 labeled 30-s epochs from 192 recordings used a leakage-controlled recording-level split that separated model fitting, early stopping, Platt calibration, threshold selection, conformal calibration, and final testing. In a five-seed controlled replication, the event-enhanced model improved the F1-score from 0.546 to 0.598, the ROC-AUC from 0.910 to 0.934, and the average precision from 0.584 to 0.645 relative to an otherwise matched Raw-Signal-only model. Paired recording-level bootstrap intervals excluded zero for the improvements in F1, ROC-AUC, average precision, accuracy, balanced accuracy, precision, specificity, and Brier score, while the recall interval included zero. Trust-based selective prediction further reduced accepted-case error by deferring lower-trust cases, although stricter acceptance thresholds also reduced accepted-case apnea recall. The Full ensemble achieved empirical conformal coverage of 0.908 at nominal 0.90. These results suggest that trust-aware decision support may provide a useful complement to prediction-centric sleep apnea screening by supporting uncertainty communication, selective abstention, and human oversight.

Index Terms—Sleep apnea screening, trustworthy artificial intelligence, explainable AI, uncertainty quantification, conformal prediction, selective prediction, clinical decision support, human-AI collaboration, trust calibration.

I. INTRODUCTION

Sleep apnea is one of the most prevalent sleep-related respiratory disorders, affecting millions of individuals worldwide and contributing to cardiovascular disease, hypertension, stroke, cognitive impairment, and reduced quality of life [1], [2]. Early identification is therefore important for timely clinical intervention and long-term patient management [2]. Polysomnography (PSG) remains the clinical gold standard for diagnosis; however, manual review and annotation of

PSG recordings are time-consuming, costly, and dependent on specialized expertise [3], [4].

Recent advances in deep learning have shown promising performance for automated sleep apnea detection from physiological signals such as airflow, thoracic movement, pulse rate, and oxygen saturation. Convolutional neural networks, recurrent neural networks, transformer-based models, and hybrid architectures have improved predictive performance across a range of sleep-analysis tasks [4]–[6]. Despite these advances, most existing approaches remain primarily prediction-centric and provide limited support for determining whether individual model outputs should be trusted in clinical practice [7], [8].

In healthcare applications, predictive performance alone is insufficient. Clinicians must consider not only the predicted label, but also the reliability of the prediction, the degree of predictive uncertainty, the consistency of supporting evidence, and whether automated output should be accepted or deferred for further review [7], [8]. Existing explainable artificial intelligence (XAI) methods can highlight influential features or temporal regions, but they rarely quantify the reliability of the generated explanations [8]–[10]. Similarly, uncertainty quantification techniques estimate predictive uncertainty but do not by themselves provide clinically actionable decision-support criteria [11]–[14].

To address these limitations, we propose **Trust-Sleep**, a trust-aware clinical decision-support framework for sleep apnea screening. Rather than treating sleep apnea detection solely as a classification problem, Trust-Sleep introduces a unified decision layer that combines predictive confidence, uncertainty estimation, conformal reliability, event-centric physiological evidence, signal quality assessment, and related decision-support factors within a single Clinical Decision Trust (CDT) score. In parallel, we introduce a Clinical Explanation Reliability Assessment (CERA) score that evaluates explanation stability, temporal consistency, physiological alignment, and related automated reliability criteria.

Building upon CDT and CERA, we further propose a Clinical Acceptance Score (CAS) to support selective prediction and clinician escalation [13], [14]. Instead of forcing a decision for every sample, the framework selectively accepts higher-trust predictions while referring lower-trust or uncertain cases for additional review. This design is intended to better reflect clinical workflows, in which reducing the risk of unreliable automated decisions may be as important as maximizing overall predictive performance.

Experimental evaluation under a recording-disjoint pediatric protocol suggests that Trust-Sleep provides useful trust-calibration and selective-prediction behavior in addition to predictive modeling. The event-enhanced Full model improved predictive performance relative to an otherwise matched Raw-

Signal-only model, while the proposed trust scores were positively associated with predictive correctness and negatively associated with predictive uncertainty. Trust-based selective prediction reduced accepted-case error by deferring lower-trust cases, although increasingly restrictive trust thresholds also reduced accepted-case apnea recall. Conformal coverage on the held-out test partition was close to the nominal target. These results nevertheless support the potential value of trust-aware decision support as a complement to conventional prediction-centric pediatric sleep apnea detection. The main contributions of this work are summarized as follows:

- We propose Trust-Sleep, a unified trust-aware framework for sleep apnea screening that jointly considers predictive modeling, reliability assessment, and clinical decision acceptance.
- We introduce Clinical Decision Trust (CDT), a trust aggregation mechanism that combines confidence, uncertainty, conformal reliability, physiological evidence, signal quality, and related decision-support factors.
- We introduce Clinical Explanation Reliability Assessment (CERA), an automated explanation-reliability measure based on stability, temporal consistency, physiological alignment, and related reliability criteria.
- We develop a selective-prediction mechanism that enables automated acceptance of higher-trust predictions and escalation of lower-trust cases for clinician review.
- We present empirical analyses showing that event-centric physiological representations improve predictive performance relative to a raw-signal-only configuration and that trust scores are associated with predictive correctness, uncertainty, and accepted-case reliability.

II. RELATED WORK

This section reviews prior work on automated sleep apnea detection, explainable and uncertainty-aware clinical AI, and selective prediction for human-supervised decision support.

A. Deep Learning for Sleep Apnea Detection

Sleep apnea has become an important application area for artificial intelligence and biomedical signal processing [4], [15]. Early automated detection systems typically relied on handcrafted features derived from physiological signals such as electrocardiography (ECG), airflow, respiratory effort, and oxygen saturation (SpO_2), followed by conventional machine learning methods including support vector machines, random forests, and gradient boosting [4], [15]. Although these approaches produced encouraging results, their performance often depended heavily on feature engineering and could be limited by variability across patient populations and recording conditions [4], [15]. The development of deep learning has substantially expanded the methodological landscape for sleep apnea detection. Convolutional neural networks (CNNs) have been widely used to learn discriminative local temporal patterns directly from physiological signals [16], [17], while recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and gated recurrent units (GRUs) have been applied to model sequential dependencies

in sleep recordings [5], [15]. More recently, transformer-based architectures have shown promise in capturing longer-range temporal relationships through self-attention mechanisms [6]. Hybrid architectures that combine convolutional, recurrent, or attention-based components have further improved predictive performance by integrating local and global temporal information [5], [6]. Despite these advances, most prior studies remain primarily prediction-centric and evaluate success mainly through classification-oriented metrics such as accuracy, sensitivity, specificity, F1-score, and ROC-AUC [6], [16], [17]. These measures are important, but they do not directly address whether an individual prediction should be trusted in a clinical setting. This limitation has motivated growing interest in explainability, uncertainty quantification, and broader trustworthy-AI methods for healthcare applications [7], [8].

A closely related foundation for the present study is the work of Frenning [18], who investigated machine-learning-based pediatric sleep-apnea detection on the Lovisenberg Dataset (LDS), the same clinical data source considered in this work. The study developed an end-to-end preprocessing and data-management framework for the LDS and systematically evaluated model architecture, preprocessing choices, and physiological signal combinations. Its strongest PSG-based classifier, using oxygen saturation (SpO_2) together with nasal airflow, achieved an accuracy of 0.94, an F1-score of 0.59, and Cohen's $\kappa = 0.56$ across the full pediatric cohort, with $\kappa = 0.69$ for patients in the severe-AHI group [18]. These results showed that clinically informative PSG signals can support competitive automated apnea-event detection even in a challenging pediatric population characterized by sparse positive events and substantial physiological variability.

The same study also examined whether contactless Somnify radar measurements could replace conventional PSG signals. Somnify-based sleep staging showed only fair-to-moderate agreement with PSG, and approximately 30% of periods scored as wake by PSG were reported as *No Signal* by the contactless device. For apnea detection, a classifier using the Somnify Relative-Distance signal together with SpO_2 achieved an F1-score of 0.35 and $\kappa = 0.32$, compared with an F1-score of 0.54 and $\kappa = 0.50$ for the corresponding PSG chest-movement and SpO_2 configuration [18]. The reduced performance was associated with the lower reliability and incomplete availability of the contactless signal, while the radar Relative-Distance measurement provides only an indirect proxy for respiratory motion. These findings motivated our use of PSG as the physiological sensing basis in the present work: before addressing whether an automated decision can be *trusted*, the underlying physiological evidence must itself be sufficiently reliable.

Importantly, however, Frenning's study primarily addressed the *predictive* question—which preprocessing strategy, signal combination, and classifier can best detect pediatric apnea events. It did not address whether an individual model prediction is sufficiently reliable for automated clinical use, whether its explanation is stable, how predictive uncertainty should be communicated, or when the model should abstain and defer a case to a clinician. The present work therefore builds on that predictive and data-processing foundation and addresses the

next decision-support gap by augmenting PSG-based apnea screening with uncertainty estimation, conformal reliability, physiological evidence, explanation-reliability assessment, and trust-aware selective prediction.

B. Explainable Artificial Intelligence in Healthcare

As AI systems become increasingly integrated into healthcare workflows, transparency and interpretability have emerged as important requirements for clinical adoption [7], [8]. Clinicians are often cautious about relying on black-box models, particularly in safety-critical environments where incorrect decisions may directly affect patient outcomes [7], [19]. Consequently, explainable artificial intelligence (XAI) has become a major research direction aimed at making model behavior more understandable to human users [7], [8].

A wide range of explanation techniques has been proposed, including SHAP, LIME, Integrated Gradients, saliency maps, attention visualizations, and counterfactual explanations [9], [10], [20]–[23]. These methods have been applied across diverse healthcare domains, including medical imaging, physiological signal analysis, disease diagnosis, and clinical risk prediction [7], [8]. Their primary goal is to identify which features, signal regions, or temporal segments contribute most strongly to a model prediction [9], [10], [20].

However, most XAI approaches focus on generating explanations rather than determining whether those explanations are themselves reliable [8], [19]. In practice, explanations may change under small input perturbations, model retraining, or distribution shift [19], [24]. As a result, clinicians may be presented with plausible-looking explanations without any direct indication of their stability, consistency, or physiological validity [19]. This limitation suggests that explanation generation alone may be insufficient for clinical decision support and that complementary mechanisms for assessing explanation reliability are needed. At the same time, explanation quality is only one aspect of trustworthiness; prediction reliability remains equally important [7], [8].

C. Uncertainty Quantification and Conformal Prediction

Uncertainty quantification has become an important component of trustworthy AI, particularly in healthcare applications where incorrect predictions may have meaningful clinical consequences [11], [25]. Unlike deterministic models that provide only point predictions, uncertainty-aware methods seek to estimate how confident a model is in its outputs and to identify cases in which predictions may be unreliable [11], [25]. Several approaches have been proposed for uncertainty estimation, including Bayesian neural networks, Monte Carlo dropout, deep ensembles, evidential learning, and related probabilistic methods [11], [25]–[27].

These techniques can provide useful information about predictive confidence and have been used to identify difficult or potentially out-of-distribution cases [11], [26]. More recently, conformal prediction has attracted substantial interest because it produces prediction sets with finite-sample coverage guarantees under stated assumptions [12], [28]. By generating

prediction sets rather than only single-label outputs, conformal methods provide a principled framework for expressing predictive uncertainty and abstention [12], [28].

Nevertheless, uncertainty estimation and conformal prediction primarily address the predictive aspect of reliability. They do not directly evaluate explanation reliability, physiological plausibility, signal quality, or the broader question of how uncertainty information should be translated into actionable clinical decisions [7], [8]. In many applications, uncertainty remains an auxiliary output rather than an integrated component of workflow-aware decision support. [25] This limitation motivates closer connections between uncertainty estimation, selective prediction, and human oversight [13], [14].

D. Selective Prediction and Human–AI Collaboration

Selective prediction has emerged as a promising strategy for deploying AI systems in high-stakes settings [13], [14]. Rather than forcing a prediction for every input, selective prediction allows a model to abstain when confidence is insufficient or estimated risk is too high [13], [29]. This approach can improve accepted-case reliability by restricting automated decisions to cases in which the model is expected to perform more safely, while deferring more difficult cases to human experts [13], [14]. In healthcare, selective prediction aligns naturally with clinical practice. Physicians routinely make decisions under uncertainty and often seek additional evidence before acting in ambiguous cases. AI systems that can recognize potential limitations and defer uncertain cases for further review may therefore be more practical and safer than systems that always issue definitive predictions. [7] Most existing selective-prediction approaches, however, rely on a single confidence score, uncertainty estimate, or risk function to determine whether a prediction should be accepted or rejected [13], [14]. Clinical trust is more naturally multidimensional. In addition to predictive confidence, clinicians may consider uncertainty, consistency of supporting evidence, physiological plausibility, explanation stability, and signal quality before accepting an automated recommendation. Existing selective-prediction frameworks rarely integrate these factors within a unified trust-aware assessment mechanism [7], [8].

E. Research Gap

The preceding discussion highlights an important gap in the current literature. Deep learning methods have improved sleep apnea detection, but most approaches remain primarily prediction-centric. Explainable AI can increase transparency, yet commonly does not assess whether generated explanations are themselves reliable. Likewise, uncertainty quantification, conformal prediction, and selective-prediction methods provide useful reliability or abstention mechanisms, but these components are often treated in isolation rather than integrated into a unified clinical decision-support process. As a result, existing approaches rarely combine predictive reliability, explanation reliability, uncertainty, physiological evidence, signal quality, and decision acceptance within a single framework [7], [8], [12], [13]. Such integration is particularly important in healthcare, where the value of an AI system depends not

only on predictive performance but also on whether its outputs can be interpreted, qualified, and acted upon safely within a human-supervised workflow [7], [19].

Prior work on the same Lovisenberg Dataset (LDS) established the feasibility of PSG-based pediatric apnea-event classification and showed that the available Somnofy measurements were not yet sufficiently reliable to replace PSG [18]. However, that work remained centered on predictive performance. The key unresolved question is therefore not only whether an apnea event can be detected, but whether an individual prediction and its supporting evidence are sufficiently reliable to justify automated acceptance or clinician review. This distinction between *predictive performance* and *decision-level trust* forms the central motivation for the present work.

To address this gap, we propose **Trust-Sleep**, a trust-aware clinical decision-support framework for sleep apnea screening. The framework integrates predictive confidence, uncertainty estimation, conformal reliability, event-centric physiological evidence, signal quality assessment, and automated explanation-reliability criteria within a unified decision-making process. Specifically, we introduce Clinical Decision Trust (CDT) to summarize prediction-related reliability factors, Clinical Explanation Reliability Assessment (CERA) to evaluate explanation stability and consistency, and a Clinical Acceptance Score (CAS) to support selective prediction and clinician escalation. By jointly modeling these complementary dimensions, Trust-Sleep extends conventional sleep apnea detection toward a more reliability-aware and human-supervised decision-support setting.

III. TRUST-SLEEP FRAMEWORK

The goal of Trust-Sleep is not only to classify sleep apnea epochs, but also to determine whether a prediction is sufficiently reliable to support clinical use [7], [8]. Conventional deep learning models typically output a predicted label and an associated confidence score. In practice, however, clinical decision support requires richer information, including predictive uncertainty, consistency of supporting evidence, explanation-related reliability, and criteria for automated acceptance or clinician review [7], [8], [25]. To address this need, we formulate sleep apnea screening as a trust-aware clinical decision-support problem.

A. Problem Definition

Consider a polysomnography (PSG) recording comprising multiple physiological signals collected during sleep. Let $\mathbf{x}_t \in \mathbb{R}^{L \times C}$ denote the signal segment associated with epoch t , where L is the temporal length of the segment and C is the number of physiological channels. In our implementation, $C = 4$ corresponding to airflow, chest movement, pulse rate, and oxygen saturation (SpO_2). For each epoch t , the objective is to predict the binary sleep apnea label $y_t \in \{0, 1\}$, where

$$y_t = \begin{cases} 0, & \text{Non-Apnea} \\ 1, & \text{Apnea.} \end{cases} \quad (1)$$

The predictive model learns a mapping $f_\theta : \mathbf{x}_t \rightarrow p_t$, where $p_t = P(y_t = 1 \mid \mathbf{x}_t)$ is the predicted probability of apnea and

$p_t \in [0, 1]$. A conventional sleep apnea detector then converts this probability into a hard decision according to

$$\hat{y}_t = \begin{cases} 1, & p_t \geq \tau \\ 0, & p_t < \tau, \end{cases} \quad (2)$$

where τ is a decision threshold. Although this formulation is sufficient for automated classification, it does not directly address several clinically relevant questions: How reliable is the prediction? How uncertain is the model? Is the supporting evidence consistent with the predicted label? Can the explanation be considered stable and plausible? Should the prediction be accepted automatically, or should it be referred for clinician review?

These limitations can be illustrated through two simple cases. In one case, an epoch may yield $p_t = 0.98$, leading a conventional model to predict apnea with high confidence. Even in this setting, however, the clinician does not know whether the prediction remains stable under perturbation, whether uncertainty is low, or whether the supporting explanation is reliable [7], [11]. In a second case, an epoch may yield $p_t = 0.54$, which may still produce the same predicted label under the threshold rule while clearly reflecting a more ambiguous decision. Treating these cases identically may reduce the practical usefulness and safety of automated decision support [13], [14].

These observations motivate a shift from prediction-centric classification toward trust-aware clinical decision support. Instead of producing only a predicted label $\hat{y}_t$, Trust-Sleep generates an enriched decision representation,

$$\mathcal{O}_t = \{\hat{y}_t, p_t, u_t, CDT_t, CERA_t, CAS_t\}, \quad (3)$$

where p_t denotes predictive probability, u_t denotes predictive uncertainty, CDT_t denotes Clinical Decision Trust, $CERA_t$ denotes Clinical Explanation Reliability Assessment, and CAS_t denotes Clinical Acceptance Score. This formulation changes the central question from “What is the predicted label?” to “How reliable is this prediction, and should it be accepted automatically or escalated for review?” From a practical perspective, the framework is intended to better reflect clinician-facing decision support, in which diagnostic labels are considered together with uncertainty, supporting evidence, and the need for additional review [7], [8], [13]. The remainder of this section introduces the Trust-Sleep architecture and then presents the mathematical formulations of Clinical Decision Trust (CDT), Clinical Explanation Reliability Assessment (CERA), Clinical Acceptance Score (CAS), conformal prediction, and selective prediction for trust-aware sleep apnea screening.

B. Trust-Sleep Architecture

Trust-Sleep is designed as a trust-aware clinical decision-support framework that combines two complementary sources of information: raw physiological dynamics and event-centric physiological descriptors. The Raw-Signal-only branch learns temporal patterns directly from PSG signals, whereas the event-centric branch summarizes clinically meaningful characteristics such as instability, slope, range, abrupt drops and

risers, and event density [4], [5]. These two representations are fused to estimate apnea risk, after which a trust layer computes predictive uncertainty, conformal reliability, Clinical Decision Trust (CDT), Clinical Explanation Reliability Assessment (CERA), and Clinical Acceptance Score (CAS). This design is motivated by the observation that clinical decisions are rarely based on waveform morphology alone. In practice, clinicians consider both the signal pattern itself and higher-level physiological evidence before deciding whether a case appears reliable or requires further review [7], [8]. Trust-Sleep follows the same principle computationally by combining learned temporal signal representations with explicit event-level descriptors and then augmenting the resulting prediction with trust-aware decision measures. Fig. 1 illustrates the overall architecture. A PSG epoch is processed in parallel by a raw physiological signal branch and an event-centric representation branch. Their outputs are fused to produce an apnea-risk probability. A downstream trust layer then derives uncertainty, conformal reliability, CDT, CERA, and CAS, which are finally used to support selective acceptance or clinician review [13], [14].

1) **Raw Physiological Signal Branch:** The raw physiological signal branch learns discriminative temporal representations directly from the PSG epoch without relying on manually engineered features. Although event-centric descriptors provide clinically interpretable summaries, they inevitably compress the original signal and may omit subtle temporal patterns that remain informative for sleep apnea detection. The purpose of the raw branch is therefore to preserve the full temporal dynamics of the physiological recordings and allow the model to discover latent signal structure automatically. For each epoch t , the input is represented as

$$\mathbf{x}_t \in \mathbb{R}^{L \times C}, \quad (4)$$

where L denotes the temporal length of the epoch and C denotes the number of physiological channels. In our implementation,

$$C = 4, \quad (5)$$

corresponding to airflow, chest movement, pulse rate, and oxygen saturation (SpO₂). Prior to convolutional processing, the input tensor is rearranged into channel-first format,

$$\tilde{\mathbf{x}}_t \in \mathbb{R}^{C \times L}, \quad (6)$$

so that one-dimensional convolutions can be applied along the temporal axis. The raw branch is modeled as

$$\mathbf{h}_t^{raw} = g_{raw}(\tilde{\mathbf{x}}_t; \theta_{raw}), \quad (7)$$

where $g_{raw}(\cdot)$ denotes the temporal convolutional encoder and θ_{raw} represents its learnable parameters. In the implemented architecture, this encoder uses three parallel temporal convolutional paths with kernel sizes 3, 7, and 15. Each path applies a $4 \rightarrow 32$ convolution, batch normalization, GELU activation, max pooling, dropout, a $32 \rightarrow 64$ convolution, batch normalization, GELU activation, and adaptive global pooling. Concatenating the three 64-dimensional path outputs yields

$$\mathbf{h}_t^{raw} \in \mathbb{R}^{d_r}, \quad (8)$$

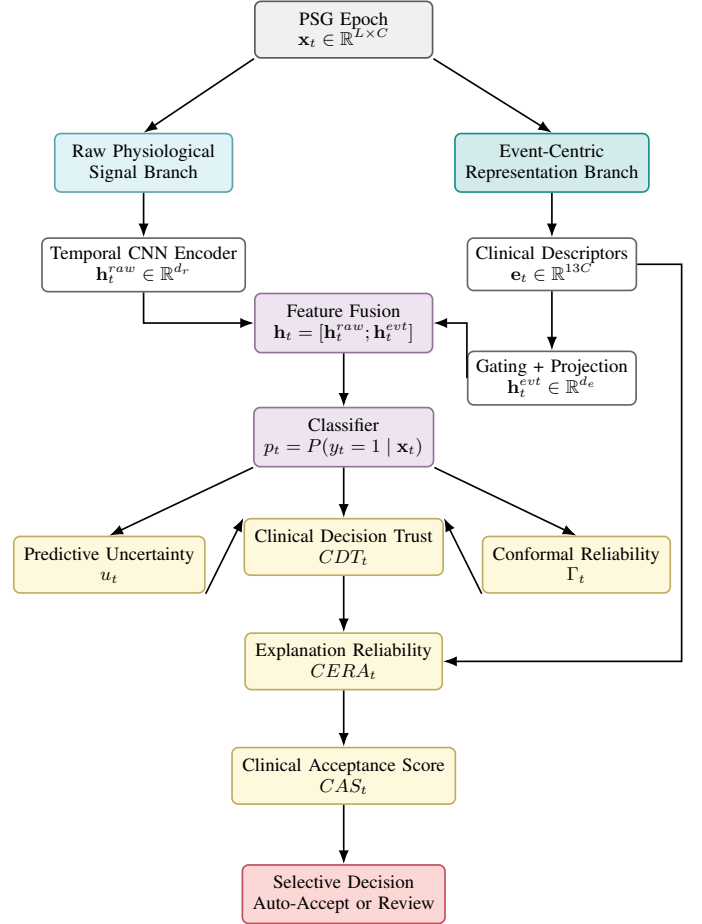


Fig. 1: Overview of the Trust-Sleep architecture. A PSG epoch is processed by a raw physiological signal branch and an event-centric representation branch. The fused representation produces an apnea-risk probability, and a trust layer then derives predictive uncertainty, conformal reliability, Clinical Decision Trust (CDT), Clinical Explanation Reliability Assessment (CERA), and Clinical Acceptance Score (CAS). The final output supports either automated acceptance or clinician review.

where, in our implementation,

$$d_r = 192. \quad (9)$$

This branch is intended to capture signal characteristics that may not be fully represented by explicit descriptors. For example, short-duration airflow fluctuations, subtle pulse variations, or mild oxygen saturation changes may jointly indicate respiratory instability even when no single handcrafted summary is sufficient on its own.

2) **Event-Centric Representation Branch:** Whereas the raw branch learns directly from waveform morphology, clinical reasoning often also relies on interpretable physiological summaries. For example, an apnea-related epoch may exhibit airflow instability, oxygen desaturation, abrupt downward or upward changes, or irregular cardiopulmonary dynamics. To represent such information explicitly, Trust-Sleep includes

an event-centric descriptor layer. For each channel $c \in \{1, \dots, C\}$, let

$$\mathbf{x}_{t,c} = [x_{t,1,c}, x_{t,2,c}, \dots, x_{t,L,c}]^\top \quad (10)$$

denote the temporal signal of channel c within epoch t . From this signal, the event-centric layer computes a collection of statistical and dynamic descriptors.

The mean and standard deviation are defined as

$$\mu_{t,c} = \frac{1}{L} \sum_{\ell=1}^L x_{t,\ell,c}, \quad (11)$$

and

$$\sigma_{t,c} = \sqrt{\frac{1}{L} \sum_{\ell=1}^L (x_{t,\ell,c} - \mu_{t,c})^2}. \quad (12)$$

The minimum, maximum, and range are given by

$$x_{t,c}^{\min} = \min_{\ell} x_{t,\ell,c}, \quad (13)$$

$$x_{t,c}^{\max} = \max_{\ell} x_{t,\ell,c}, \quad (14)$$

and

$$r_{t,c} = x_{t,c}^{\max} - x_{t,c}^{\min}. \quad (15)$$

To characterize temporal changes, we define first-order differences

$$\Delta x_{t,\ell,c} = x_{t,\ell,c} - x_{t,\ell-1,c}, \quad \ell = 2, \dots, L. \quad (16)$$

The average slope and slope variability are then computed as

$$m_{t,c} = \frac{1}{L-1} \sum_{\ell=2}^L \Delta x_{t,\ell,c}, \quad (17)$$

and

$$s_{t,c} = \sqrt{\frac{1}{L-1} \sum_{\ell=2}^L (\Delta x_{t,\ell,c} - m_{t,c})^2}. \quad (18)$$

Abupt downward and upward changes are summarized by

$$d_{t,c}^- = \frac{1}{L-1} \sum_{\ell=2}^L \max(-\Delta x_{t,\ell,c}, 0), \quad (19)$$

and

$$d_{t,c}^+ = \frac{1}{L-1} \sum_{\ell=2}^L \max(\Delta x_{t,\ell,c}, 0). \quad (20)$$

The instability score is defined as

$$\iota_{t,c} = \frac{s_{t,c}}{\sigma_{t,c} + \epsilon}, \quad (21)$$

where ϵ is a small constant introduced to avoid division by zero. Finally, event density measures the fraction of temporal changes whose magnitude exceeds the local slope variability:

$$\rho_{t,c} = \frac{1}{L-1} \sum_{\ell=2}^L \mathbb{I}(|\Delta x_{t,\ell,c}| > s_{t,c}), \quad (22)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. For each channel, the descriptor vector includes mean, standard deviation, minimum, maximum, range, slope statistics, positive and negative

changes, instability, tail-pressure descriptors, and event density. Each channel therefore contributes 13 descriptors. The full event representation is

$$\mathbf{e}_t = [\mathbf{e}_{t,1}; \mathbf{e}_{t,2}; \dots; \mathbf{e}_{t,C}] \in \mathbb{R}^{13C}. \quad (23)$$

Since $C = 4$ in our implementation, the resulting descriptor dimension is

$$13C = 52. \quad (24)$$

Before projection, the 52-dimensional descriptor vector is passed through a learned gate,

$$\mathbf{g}_t = \sigma(W_g \text{LN}(\mathbf{e}_t) + \mathbf{b}_g), \quad \tilde{\mathbf{e}}_t = \mathbf{e}_t \odot \mathbf{g}_t, \quad (25)$$

which allows the model to modulate the contribution of individual event descriptors. The gated descriptor vector is then passed through a nonlinear projection network,

$$\mathbf{h}_t^{evt} = g_{evt}(\tilde{\mathbf{e}}_t; \theta_{evt}), \quad (26)$$

where

$$\mathbf{h}_t^{evt} \in \mathbb{R}^{d_e}. \quad (27)$$

In our implementation,

$$d_e = 80. \quad (28)$$

The role of this branch is to provide explicit physiological summaries that complement the raw waveform representation. For brevity, only the principal descriptors are shown explicitly above; the implemented descriptor set additionally includes tail-pressure statistics used in the final 13-descriptor representation per channel.

3) Feature Fusion and Classification: The raw and event-centric branches encode complementary information: the raw branch captures latent temporal patterns, whereas the event-centric branch captures structured physiological evidence. Trust-Sleep combines these representations by concatenation,

$$\mathbf{h}_t = [\mathbf{h}_t^{raw}; \mathbf{h}_t^{evt}]. \quad (29)$$

Given $\mathbf{h}_t^{raw} \in \mathbb{R}^{192}$ and $\mathbf{h}_t^{evt} \in \mathbb{R}^{80}$, the fused representation has dimension

$$\mathbf{h}_t \in \mathbb{R}^{272}. \quad (30)$$

The classifier head uses hidden dimensions $256 \rightarrow 128$ before the final binary logit. The fused representation is passed through a classifier head,

$$z_t = g_{cls}(\mathbf{h}_t; \theta_{cls}), \quad (31)$$

where z_t is the apnea-risk logit. The probability of apnea is then obtained through the sigmoid function:

$$p_t = \sigma(z_t) = \frac{1}{1 + \exp(-z_t)}. \quad (32)$$

The predicted class is determined using the operating threshold τ :

$$\hat{y}_t = \begin{cases} 1, & p_t \geq \tau, \\ 0, & p_t < \tau. \end{cases} \quad (33)$$

Unlike a conventional classifier, however, the predicted label $\hat{y}_t$ is not the final output of Trust-Sleep. Instead, the fused

representation, event descriptors, predicted probability, uncertainty estimates, and conformal prediction set are passed to the trust layer. This layer computes CDT, CERA, and CAS, which are then used to determine whether the prediction should be automatically accepted or referred for clinician review. Table I summarizes the major representations in the architecture.

TABLE I: Major representations in the Trust-Sleep architecture.

Symbol	Description	Dimension
$\mathbf{x}_t$	PSG epoch	$L \times C$
$\tilde{\mathbf{x}}_t$	Channel-first PSG epoch	$C \times L$
$\mathbf{h}_t^{raw}$	Raw temporal feature vector	192
$\mathbf{e}_t$	Event descriptor vector	13C
$\mathbf{h}_t^{evt}$	Event feature vector	80
$\mathbf{h}_t$	Fused representation	272
z_t	Apnea-risk logit	1
p_t	Apnea-risk probability	1

Running clinical example. Suppose an epoch shows irregular airflow, increased chest movement instability, and a moderate decrease in SpO_2 . The raw signal branch may detect latent temporal changes in the waveform, while the event-centric branch explicitly captures descriptors such as airflow instability, event density, and oxygen saturation range. If the classifier outputs $p_t = 0.86$, a conventional system would simply predict apnea. Trust-Sleep, however, continues by assessing whether this decision is sufficiently reliable for automatic acceptance. If uncertainty is low, the conformal prediction set is singleton, the physiological evidence is consistent, and the explanation is stable, the framework may assign high CDT and CAS values and accept the prediction automatically. Conversely, if $p_t = 0.86$ but uncertainty is high or the explanation appears unstable, the same apnea prediction may be escalated for clinician review. The architecture therefore separates two clinically distinct tasks: estimating apnea risk and determining whether the resulting prediction is trustworthy enough for automated acceptance.

C. Clinical Decision Trust (CDT)

Prediction confidence alone is insufficient for clinical decision support. A model may produce a highly confident prediction while operating in an unfamiliar region of the feature space, under poor signal quality, or in the presence of weak or inconsistent physiological evidence. A more useful clinical decision measure should therefore reflect not only the predicted probability, but also the reliability of the evidence supporting that prediction. To address this need, we define *Clinical Decision Trust* (CDT), a multidimensional trust score that summarizes several complementary sources of prediction reliability. In the general Trust-Sleep formulation, CDT combines predictive confidence, uncertainty reliability, conformal reliability, physiological evidence strength, signal quality, and robustness-related factors. Let the component scores be denoted by

- C_t : predictive confidence,
- U_t : uncertainty reliability,
- R_t : conformal reliability,

- E_t : physiological evidence strength,
- Q_t : signal quality, and
- P_t : robustness-related reliability.

The predictive confidence component is obtained directly from the classifier output probability p_t . Predictions closer to either 0 or 1 indicate stronger confidence than predictions near the decision boundary. Accordingly, we define,

$$C_t = \max(p_t, 1 - p_t). \quad (34)$$

Predictive uncertainty is estimated using Monte Carlo dropout at inference time. Let $p_t^{(m)}$ denote the predicted probability from stochastic forward pass m , and let

$$\bar{p}_t = \frac{1}{M} \sum_{m=1}^M p_t^{(m)} \quad (35)$$

be the mean prediction across M stochastic passes. The predictive variance is then

$$u_t = \frac{1}{M} \sum_{m=1}^M \left(p_t^{(m)} - \bar{p}_t \right)^2. \quad (36)$$

Because lower uncertainty should correspond to higher trust, uncertainty reliability is defined through normalized inversion:

$$U_t = 1 - \frac{u_t - u_{\min}}{u_{\max} - u_{\min}}. \quad (37)$$

Conformal prediction is used to quantify statistical reliability. Let α_t denote the nonconformity score associated with epoch t . We define conformal reliability as

$$R_t = 1 - \alpha_t. \quad (38)$$

The physiological evidence component summarizes the strength of clinically relevant evidence extracted from the event-centric representation. Let $e_{t,k}^{\text{norm}}$ denote the normalized value of descriptor k . Then

$$E_t = \frac{1}{K} \sum_{k=1}^K e_{t,k}^{\text{norm}}, \quad (39)$$

where K is the total number of event descriptors. Signal quality reflects the integrity of the physiological recordings and penalizes heavily corrupted epochs. Let η_t denote the estimated artifact or noise level. Signal-quality reliability is defined as

$$Q_t = 1 - \frac{\eta_t}{\eta_{\max}}. \quad (40)$$

Finally, the framework can include a robustness-related component that evaluates whether predictions remain stable under designated perturbations or protected-processing conditions. Let p_t^{rob} denote the corresponding prediction under such perturbation. A simple robustness reliability term is

$$P_t = 1 - |p_t - p_t^{rob}|. \quad (41)$$

The final CDT score is obtained through weighted aggregation:

$$CDT_t = w_C C_t + w_U U_t + w_R R_t + w_E E_t + w_Q Q_t + w_P P_t, \quad (42)$$

subject to,

$$w_C + w_U + w_R + w_E + w_Q + w_P = 1. \quad (43)$$

Unless otherwise specified, equal weighting may be used as an initial default:

$$w_C = w_U = w_R = w_E = w_Q = w_P = \frac{1}{6}. \quad (44)$$

For the leakage-controlled empirical evaluation reported in this paper, the perturbation term was analyzed separately and the operational CDT instantiation used

$$CDT_t = 0.30C_t + 0.25U_t + 0.20R_t + 0.15E_t + 0.10Q_t. \quad (45)$$

All component quantities are normalized and clipped to $[0, 1]$ before aggregation. By construction, $CDT_t \in [0, 1]$. A high CDT score indicates that the prediction is supported by strong confidence, low uncertainty, reliable conformal behavior, coherent physiological evidence, and acceptable signal quality. A low CDT score indicates that the prediction should be interpreted more cautiously and may be more appropriate for clinician review.

D. Clinical Explanation Reliability Assessment (CERA)

Prediction reliability alone is not sufficient for trustworthy decision support if the accompanying explanation is unstable, inconsistent, or physiologically implausible. We therefore define Clinical Explanation Reliability Assessment (CERA) as a general framework for aggregating complementary indicators of explanation reliability. In the general formulation presented here, CERA comprises four operational dimensions: perturbation stability, temporal consistency, reliability consistency, and physiological plausibility. The Experiment 1 implementation additionally included an automated interpretability heuristic, denoted by H_t . This component was designed as a model-derived proxy for the concentration and apparent interpretability of the explanation; it was computed automatically and did not use ratings from clinicians or other human evaluators. Therefore, H_t should be interpreted only as an automated explanation-reliability heuristic, not as a clinically validated measure of human interpretability. Let these components be denoted by,

- S_t : explanation stability,
- T_t : temporal consistency,
- R_t^{exp} : reliability consistency, and
- P_t^{phys} : physiological plausibility.

Explanation stability measures sensitivity to small input perturbations. Let ϕ_t denote the explanation vector for epoch t , and let ϕ_t^{pert} denote the corresponding explanation after a small perturbation. We define

$$S_t = \max \left\{ 0, 1 - \frac{\|\phi_t - \phi_t^{\text{pert}}\|_2}{\max_{\tau} \|\phi_{\tau}\|_2 + \varepsilon} \right\}, \quad (46)$$

where $\varepsilon > 0$ is a small constant introduced for numerical stability. Temporal consistency measures whether neighboring epochs yield coherent explanations:

$$T_t = \max \left\{ 0, 1 - \frac{\|\phi_t - \phi_{t-1}\|_2}{\max_{\tau} \|\phi_{\tau}\|_2 + \varepsilon} \right\}. \quad (47)$$

Reliability consistency measures agreement between explanation-related behavior and prediction-level trust:

$$R_t^{\text{exp}} = 1 - |CDT_t - C_t|. \quad (48)$$

Physiological plausibility evaluates whether explanation importance is concentrated on clinically meaningful variables or regions. Let Ω denote the set of physiologically relevant features. We define

$$P_t^{\text{phys}} = \frac{\sum_{i \in \Omega} |\phi_{t,i}|}{\sum_i |\phi_{t,i}| + \varepsilon}. \quad (49)$$

The final CERA score is obtained by weighted aggregation:

$$CERA_t = v_S S_t + v_T T_t + v_R R_t^{\text{exp}} + v_P P_t^{\text{phys}}, \quad (50)$$

$$v_S + v_T + v_R + v_P = 1.$$

With equal weighting,

$$v_S = v_T = v_R = v_P = \frac{1}{4}, \quad CERA_t \in [0, 1]. \quad (51)$$

For reproducibility, the Experiment 1 implementation additionally included an automated interpretability heuristic, denoted by H_t . This component was computed automatically and did not use ratings from clinicians or other human evaluators.

$$CERA_t^{(E1)} = \frac{S_t + T_t + R_t^{\text{exp}} + P_t^{\text{phys}} + H_t}{5}. \quad (52)$$

All component scores, including H_t in Experiment 1, are normalized or clipped $[0, 1]$ before aggregation, where H_t denotes an automated heuristic rather than a clinician-derived interpretability score. High CERA values indicate explanations that are stable under perturbation, consistent across neighboring epochs, aligned with prediction-level trust, and concentrated on physiologically relevant features. Low CERA values indicate that an explanation may be unreliable even when the corresponding prediction appears confident. Together, CDT and CERA provide complementary assessments of prediction-level and explanation-level reliability. These two quantities form the basis of the Clinical Acceptance Score (CAS), introduced in the following subsection, which determines whether a prediction should be automatically accepted or escalated for clinician review.

E. Clinical Acceptance Score (CAS)

Whereas CDT summarizes prediction-level reliability and CERA evaluates explanation-level reliability, clinical decision support ultimately requires a unified criterion that considers both. A prediction should not be accepted automatically merely because it is confident if the associated explanation is unstable or physiologically inconsistent. Conversely, an apparently reliable explanation does not by itself guarantee that the underlying prediction is sufficiently trustworthy. To combine these two aspects, we define the *Clinical Acceptance Score* (CAS). CAS serves as the final trust-aware score used to determine whether a prediction is suitable for automatic acceptance or should instead be escalated for clinician review. The score is defined as a weighted combination of CDT and CERA:

$$CAS_t = \lambda_{CDT} CDT_t + \lambda_{CERA} CERA_t, \quad (53)$$

subject to

$$\lambda_{CDT} + \lambda_{CERA} = 1. \quad (54)$$

Because incorrect predictions generally present greater clinical risk than imperfect explanations, slightly greater weight is assigned to prediction-level trust. Accordingly, we use

$$\lambda_{CDT} = 0.60, \quad \lambda_{CERA} = 0.40. \quad (55)$$

Substituting these values into Eq. (53) yields

$$CAS_t = 0.60 CDT_t + 0.40 CERA_t. \quad (56)$$

By construction,

$$CAS_t \in [0, 1]. \quad (57)$$

High CAS values indicate that both the prediction and its supporting explanation satisfy the framework's reliability criteria, whereas low CAS values indicate that one or both components should be treated with caution. Two representative examples illustrate the behavior of CAS.

Case 1: High prediction trust and high explanation reliability.

$$CDT_t = 0.94, \quad CERA_t = 0.91. \quad (58)$$

The resulting acceptance score is

$$CAS_t = 0.60(0.94) + 0.40(0.91) = 0.928. \quad (59)$$

Such a case would generally be considered suitable for automatic acceptance.

Case 2: High prediction trust but low explanation reliability.

$$CDT_t = 0.90, \quad CERA_t = 0.42. \quad (60)$$

The corresponding score becomes

$$CAS_t = 0.60(0.90) + 0.40(0.42) = 0.708. \quad (61)$$

Although the prediction-level trust remains relatively high, the low explanation-reliability score reduces the overall acceptance score and may justify additional clinical review. CAS therefore provides a single operational decision criterion that jointly reflects prediction-related and explanation-related reliability.

F. Conformal Prediction Layer

Predicted probabilities provide useful information about model confidence, but they do not by themselves supply formal statistical coverage control. In high-stakes clinical applications, it is therefore useful to complement probability estimates with a principled mechanism for identifying uncertain cases. To this end, Trust-Sleep incorporates a conformal prediction layer that produces prediction sets rather than only single-label outputs. Let,

$$\mathcal{D}_{cal} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_{cal}} \quad (62)$$

denote the calibration dataset used after model training. For each calibration sample, a nonconformity score is computed as,

$$\alpha_i = 1 - p_i(y_i), \quad (63)$$

where $p_i(y_i)$ is the predicted probability assigned to the true label. Given a target confidence level $1 - \epsilon$, the conformal

threshold is defined as the empirical $(1 - \epsilon)$ quantile of the calibration nonconformity scores:

$$q_\epsilon = Q_{1-\epsilon} \left(\{\alpha_i\}_{i=1}^{N_{cal}} \right). \quad (64)$$

For a new sample, a candidate label y is included in the conformal prediction set if $1 - p_t(y) \leq q_\epsilon$. The resulting conformal prediction set is therefore

$$\Gamma_t = \{y : 1 - p_t(y) \leq q_\epsilon\}. \quad (65)$$

The size of Γ_t provides an interpretable indication of predictive certainty. A singleton set, $|\Gamma_t| = 1$, indicates that only one class is retained at the specified confidence level, whereas larger or empty sets indicate greater uncertainty or mismatch with the calibration distribution. In Trust-Sleep, this conformal information is incorporated into CDT through the conformal reliability component R_t . From a clinical perspective, the conformal layer provides a principled abstention-related mechanism for identifying cases that may warrant additional caution. Although conformal methods offer finite-sample coverage guarantees under their standard assumptions, empirical coverage may still degrade under distribution shift, and therefore the resulting prediction sets should be interpreted as decision-support signals rather than unconditional guarantees in external deployment.

G. Selective Prediction and Clinical Escalation

A central objective of Trust-Sleep is not only to generate predictions, but also to determine when those predictions are sufficiently reliable for automated use. To this end, the framework adopts a selective-prediction strategy that accepts higher-trust predictions while escalating lower-trust cases for clinician review. Conventional classifiers produce a decision for every sample regardless of uncertainty or reliability. By contrast, selective prediction introduces a reject option that allows the model to abstain when trust is insufficient. Let τ_{CAS} be the acceptance threshold applied to the Clinical Acceptance Score. The resulting decision policy is

$$D_t = \begin{cases} \text{Accept,} & CAS_t \geq \tau_{CAS}, \\ \text{Review,} & CAS_t < \tau_{CAS}, \end{cases} \quad (66)$$

where epochs assigned to *Accept* are considered suitable for automated decision support, whereas epochs assigned to *Review* are referred for human evaluation. The overall workflow can be summarized conceptually as,

$$\mathbf{x}_t \rightarrow \hat{y}_t \rightarrow CDT_t \rightarrow CERA_t \rightarrow CAS_t \rightarrow D_t. \quad (67)$$

This formulation supports human-AI collaboration by using trust estimates to prioritize cases for automation or review rather than treating all predictions identically. As a simple example, consider three epochs with,

$$CAS_1 = 0.95, \quad CAS_2 = 0.74, \quad CAS_3 = 0.42. \quad (68)$$

Using an acceptance threshold of $\tau_{CAS} = 0.80$,

the first prediction would be accepted automatically, whereas the second and third would be escalated for clinician review. This selective-prediction policy provides two practical

advantages. First, it can reduce the risk of incorrect automated decisions by abstaining on cases with insufficient trust. Second, it allows clinicians to focus their attention on ambiguous or difficult cases while permitting more reliable cases to be handled more efficiently. In this way, Trust-Sleep extends sleep apnea screening from a conventional classification task toward a trust-aware clinical decision-support workflow that explicitly balances automation and human oversight.

IV. EXPERIMENTAL SETUP

This section describes the two-experiment study design, the corresponding research questions, the pediatric cohort, the data and preprocessing procedures, and the evaluation protocols used to distinguish framework-level trust analyses from the controlled predictive replication.

A. Pediatric Cohort and Analysis Dataset

The source Lovisenberg Dataset (LDS) contains sleep studies from 195 pediatric patients aged 1–12 years, including 111 male and 84 female patients [18]. All source studies are overnight recordings acquired in a specialist sleep-clinic setting and scored by a sleep specialist. The source dataset contains PSG recordings accompanied by manually scored sleep stages and respiratory-event annotations, including obstructive, central, mixed, desaturation, and hypopnea events. There is no healthy control group in the source cohort; all patients had underlying disorders [18]. The source dataset also contains contactless Somnofy measurements, but those signals were not used in the present four-channel PSG analysis.

The pediatric setting is important for interpretation of the study. The age range spans substantial developmental variation in respiratory rate and physiology, and pediatric scoring criteria are not interchangeable with adult scoring criteria. In addition, sensor placement and overnight signal acquisition can be more difficult in children: movement, sensor intolerance, or sensor removal may result in weak, noisy, or missing segments [18]. The youngest participants in the source cohort were one year old; the cohort does not include infants younger than one year. These characteristics motivate reliability-aware screening and also mean that the results should be interpreted specifically within a pediatric population.

The analysis export contains 240,043 labeled 30-s epochs from 192 recording identifiers. Each epoch is represented as a 30×4 tensor containing thoracic movement, airflow, pulse rate, and oxygen saturation (SpO_2). The export contains 13,947 positive and 226,096 negative epochs, corresponding to a positive prevalence of 5.81%. The difference between the 195-patient source cohort and the 192-recording analysis export reflects the finalized analysis material available to the present pipeline; we therefore report the source-cohort demographics separately from the recording counts used for model evaluation.

B. Two-Experiment Study Design

The empirical study was organized as two complementary experiments.

TABLE II: Pediatric source cohort and finalized analysis export.

Property	Value
Source cohort	195 pediatric patients
Age range	1–12 years
Sex	111 male, 84 female
Setting	Overnight specialist sleep clinic
Healthy control group	None
Clinical population	Underlying disorders present
Analysis export	192 recording identifiers
Analysis epochs	240,043 (30 s each)
Positive / negative epochs	13,947 / 226,096
Analysis channels	Chest, airflow, pulse, SpO_2

Experiment 1 evaluates the complete Trust-Sleep trust-aware framework using a prespecified Full-model seed (271828). Its purpose is framework-level evaluation of predictive performance, Clinical Decision Trust (CDT), Clinical Explanation Reliability Assessment (CERA), Clinical Acceptance Score (CAS), uncertainty analysis, selective prediction, and perturbation robustness.

Experiment 2 provides a stricter five-seed robustness replication and controlled predictive ablation. Its purpose is to quantify variability across random initialization and to isolate whether the gated event-centric representation improves performance relative to an otherwise matched Raw-Signal-only model. Full and Raw-Signal-only configurations use the same frozen recording-level split, preprocessing, optimization, early stopping, probability calibration, threshold-selection, conformal-calibration, and final-test partitions. Five prespecified seeds are used, and Full-versus-Raw differences are evaluated using paired recording-level bootstrap intervals.

The two experiments address different questions and are therefore interpreted separately rather than pooled. Experiment 1 evaluates the functionality of the end-to-end trust framework, whereas Experiment 2 evaluates the reproducibility of predictive performance and the incremental predictive contribution of the event branch. All experiments used a fixed recording-level partition of the pediatric dataset, with mutually exclusive subsets for training, validation, probability calibration, threshold selection, conformal calibration, and final testing. All epochs from a given recording were assigned to only one subset, preventing recording-level data leakage. The held-out test subset was accessed only after preprocessing, model fitting, model selection, and calibration procedures had been finalized. Table III summarizes the purpose and principal limitation of each stage.

C. Research Questions

Because the two experiments address different forms of evidence, their research questions are presented separately. Experiment 1 is organized around RQ1-RQ4, while the Experiment 2 questions are labeled E2-Q1 to E2-Q4.

1) *Experiment 1 Research Questions: RQ1: Does Clinical Decision Trust correlate with prediction correctness?* For epoch t , correctness was defined as,

$$\text{Correct}_t = \mathbb{I}(\hat{y}_t = y_t), \quad (69)$$

TABLE III: Purpose and interpretation of the two experimental stages.

Stage	Primary purpose	Main analyses	Principal limitation
Experiment 1: Trust-Sleep framework	Evaluation of the complete trust-aware framework	Prediction, CDT, CERA, CAS, uncertainty, selective prediction, and perturbation robustness	Single prespecified fitted model for trust analyses
Experiment 2: five-seed replication	Reproducibility assessment and controlled Full-versus-Raw-Signal-only ablation	Five independent seeds, calibration, conformal analysis, ensembling, and paired recording bootstrap	Same single-center pediatric cohort; no independent-site validation

and the association was evaluated using Pearson and Spearman correlations between CDT_t and $Correct_t$.

RQ2: Do higher trust scores correspond to lower predictive uncertainty? This question was evaluated using Pearson and Spearman correlations between CDT/CAS and the uncertainty estimate derived from MC-dropout inference. A negative correlation indicates that uncertainty increases as the reported trust score decreases.

RQ3: Are prediction errors concentrated among low-trust samples? Predictions were stratified into low ($CDT < 0.60$), medium ($0.60 \leq CDT < 0.80$), and high ($CDT \geq 0.80$) groups, and the group-specific error rate was defined as

$$ErrorRate_g = \frac{FP_g + FN_g}{N_g}. \quad (70)$$

RQ4: Does selective prediction improve reliability? Acceptance was evaluated using CDT thresholds. At threshold τ , coverage was defined as

$$Coverage(\tau) = \frac{N_{accepted}(\tau)}{N_{total}}. \quad (71)$$

Accuracy, precision, recall, and F1-score were recomputed on the accepted subset to characterize the coverage–reliability trade-off. Experiment 1 additionally reports predictive baseline comparisons, CERA component behavior, and perturbation robustness. Although CAS serves as the general framework-level score for combining prediction- and explanation-related trust, the Experiment 1 selective- prediction analysis applies thresholds directly to CDT. This isolates the behavior of prediction-level trust when determining which cases are retained for automatic acceptance. CAS is evaluated separately through its associations with prediction correctness and predictive uncertainty.

2) *Experiment 2 Research Questions:* **E2-Q1: Does the event-centric branch improve predictive performance relative to an otherwise matched Raw-Signal-only model?** This question was evaluated using Full-minus-Raw differences in accuracy, balanced accuracy, precision, recall, specificity, F1-score, ROC-AUC, average precision, and Brier score, together with paired recording-level bootstrap confidence intervals.

E2-Q2: Are Full-model results stable across random initialization? This question was evaluated using the mean and standard deviation of the five independently trained seed-level models and by comparing between-seed variability for Full and Raw-Signal-only.

E2-Q3: What predictive and calibration performance is achieved by the frozen five-seed logit ensemble? The frozen five-seed ensemble was evaluated using discrimination, threshold-dependent performance, Brier score, 15-bin expected calibration error (ECE), and the confusion matrix. Ensemble

results are reported separately from seed-level means and standard deviations.

E2-Q4: Does conformal calibration retain its nominal behavior on the held-out recording-disjoint test partition?

At nominal $\alpha = 0.10$, the analysis reports empirical epoch-level coverage together with singleton, ambiguous-set, and empty-set rates. This is an empirical assessment on a held-out partition from the same pediatric cohort rather than an external-site coverage guarantee.

D. Leakage-Controlled Data Partitioning

The finalized 240,043-epoch export was treated as the single master analysis dataset. The 192 recording identifiers were partitioned once at recording level into mutually exclusive subsets for model fitting, early stopping, probability calibration, operating-threshold selection, conformal calibration, and final testing. Candidate group-exclusive partitions were searched to approximate the target fractions while retaining both classes. No recording identifier appears in more than one subset.

E. Preprocessing and Model Configurations

Missing or non-finite channel values were retained until partitioning and then imputed using channel means estimated from the training subset only. Channel-wise means and standard deviations were likewise estimated using the training subset only and frozen before transformation of validation, calibration, and test data. The same preprocessing parameters were used for Full and Raw-Signal-only models.

The Full model used the multi-path raw encoder described in Section III-B together with the 52-dimensional gated event-centric branch. The raw encoder uses parallel kernels 3, 7, and 15, producing a 192-dimensional raw representation. The gated event branch produces an 80-dimensional representation, yielding a 272-dimensional fused vector followed by a $272 \rightarrow 256 \rightarrow 128 \rightarrow 1$ classifier. Raw-Signal-only removes the descriptor/gating/event branch and uses the same raw encoder followed by a $192 \rightarrow 256 \rightarrow 128 \rightarrow 1$ classifier. Thus, the intended predictive difference between Full and Raw-Signal-only is the event-centric branch.

F. Training, Calibration, and Trust Evaluation

Both configurations were trained with AdamW, learning rate 10^{-3} , weight decay 10^{-4} , batch size 1,024, dropout 0.35, binary cross-entropy loss, and a maximum of 20 epochs. Early stopping used validation binary cross-entropy with patience 5. Five prespecified seeds were used in Experiment 2: {271828, 314159, 161803, 141421, 173205}. Experiment 1 used the Full model corresponding to seed 271828.

TABLE IV: Research questions, objectives, and corresponding evaluation evidence for the two experiments.

Question	Research objective	Evaluation evidence
RQ1	Association between CDT and prediction correctness	Pearson and Spearman correlations with $Correct_t$
RQ2	Association between trust and predictive uncertainty	Pearson and Spearman correlations with MC-dropout uncertainty
RQ3	Concentration of errors in low-trust cases	Error rate within low-, medium-, and high-CDT groups
RQ4	Reliability improvement through selective prediction	Coverage and accepted-case accuracy, precision, recall, and F1-score
E2-Q1	Incremental predictive contribution of event features	Paired Full-minus-Raw recording-bootstrap differences
E2-Q2	Stability across model initialization	Five-seed mean, standard deviation, and between-seed comparison
E2-Q3	Performance of the frozen five-seed ensemble	Discrimination, operating-point, calibration, and confusion metrics
E2-Q4	Held-out behavior of conformal prediction	Empirical coverage and prediction-set-size rates

TABLE V: Frozen recording-level partition used by both experiments.

Split	Recordings	Epochs	Positive	Negative	Positive rate
Training	115	144,038	8,371	135,667	5.81%
Early-stopping validation	20	24,092	1,396	22,696	5.79%
Platt calibration	10	12,013	678	11,335	5.64%
Threshold selection	10	12,105	689	11,416	5.69%
Conformal calibration	18	23,839	1,413	22,426	5.93%
Final test	19	23,956	1,400	22,556	5.84%

The same frozen recording split was used for every seed and both configurations.

Probability calibration used Platt scaling fit only on the dedicated 5% Platt partition. The operating threshold was selected only on the separate 5% threshold-selection partition by maximizing F1-score, with the higher threshold used to break ties. Split-conformal calibration used only the dedicated 10% conformal partition at $\alpha = 0.10$. The final 10% test partition was accessed only after preprocessing, model selection, probability calibration, threshold selection, and conformal calibration had been fixed. For Experiment 1, MC dropout used 15 stochastic forward passes. The operational trust score was,

$$CDT_t = 0.30C_t + 0.25U_t + 0.20R_t + 0.15E_t + 0.10Q_t, \quad (72)$$

where the terms denote predictive confidence, uncertainty reliability, conformal reliability, physiological evidence, and signal quality. CERA used the five-component Experiment 1 definition introduced earlier,

$$CERA_t = \frac{S_t + T_t + R_t^{\text{exp}} + P_t^{\text{phys}} + H_t}{5}, \quad (73)$$

and the empirical Clinical Acceptance Score was

$$CAS_t = 0.60CDT_t + 0.40CERA_t. \quad (74)$$

The automated H_t term is a concentration heuristic rather than a clinician-rated interpretability measure. Laplace perturbations parameterized by $\epsilon \in \{0.5, 1, 2, 5, 10\}$ were evaluated separately as a perturbation-robustness analysis and are not presented as a formal differential-privacy mechanism.

G. Evaluation Metrics and Statistical Analysis

Predictive evaluation included accuracy, balanced accuracy, precision, recall, specificity, F1-score, ROC-AUC, average precision, Brier score, and 15-bin ECE. Variation across random initialization was summarized as the mean and sample standard deviation across the five independently trained models.

Final ensemble predictions were produced by averaging the five pre-sigmoid logits and then fitting/applying the ensemble Platt calibrator using only the dedicated calibration partition.

Confidence intervals were estimated using a paired recording-level nonparametric bootstrap. Complete test recordings, rather than individual epochs, were sampled with replacement to preserve within-recording dependence. Full-minus-Raw differences were computed within each paired bootstrap sample, and percentile 95% confidence intervals were obtained from 1,000 repetitions.

V. RESULTS AND ANALYSIS

This section reports the two experiments defined in the preceding Experimental Setup. Experiment 1 evaluates the Trust-Sleep framework using the prespecified Full-model seed, including predictive performance, trust-related analyses, selective prediction, perturbation robustness, and baseline comparisons. Experiment 2 then reports the five-seed Full-versus-Raw-Signal-only ablation. Both analyses use the same leakage-controlled recording-level partition, but they address different evidentiary questions and are reported separately.

A. Experiment 1: Predictive Performance

Table VI summarizes the predictive performance of Trust-Sleep on the held-out recording-disjoint pediatric test partition. Trust-Sleep achieved an overall accuracy of 95.6%, a balanced

TABLE VI: Predictive performance of Trust-Sleep on the held-out pediatric test partition.

Metric	Value	Metric	Value
Accuracy	0.956	Specificity	0.982
Balanced Accuracy	0.760	F1-score	0.588
Precision	0.649	ROC-AUC	0.928
Recall	0.538	Average Precision	0.619
Brier score	0.0336	ECE	0.0075

accuracy of 76.0%, and a ROC-AUC of 0.928. Given the substantial class imbalance, discrimination-oriented metrics such

as ROC-AUC, average precision, recall, balanced accuracy, and F1-score are more informative than raw accuracy alone. The recall of 53.8% indicates that the model identified a meaningful proportion of apnea epochs, although a substantial fraction of positive epochs remained missed. The held-out confusion matrix contained 22,149 true negatives, 407 false positives, 647 false negatives, and 753 true positives. These results support useful epoch-level ranking and moderate threshold-dependent apnea detection while also motivating the trust-aware review layer rather than autonomous exclusion of disease.

B. Experiment 1: Trust Validation

A central objective of Trust-Sleep is not only to detect apnea epochs, but also to provide useful indicators of prediction reliability. To assess whether CDT, CERA, and CAS behaved consistently with this objective, Experiment 1 evaluated their associations with prediction correctness, predictive uncertainty, and selective-prediction behavior.

1) *Correlation Between Trust and Correctness*: To address RQ1, correlations between the proposed trust measures and prediction correctness were computed and are reported in Table VII. All three trust measures exhibited positive associations

TABLE VII: Correlation between trust measures and prediction correctness.

Metric	Pearson	Spearman
CDT vs Correctness	0.378	0.198
CERA vs Correctness	0.156	0.154
CAS vs Correctness	0.391	0.216

with prediction correctness, with CAS showing the strongest Pearson association (0.391), closely followed by CDT (0.378). Mean CDT was 0.848 for correct predictions and 0.669 for incorrect predictions, while mean CAS was 0.802 and 0.685, respectively. These associations do not establish causal or clinical validity, but they indicate that the operational trust measures stratify decision quality beyond the predicted label alone.

2) *Correlation Between Trust and Uncertainty*: To address RQ2, Pearson and Spearman correlations were computed between the proposed trust measures and predictive uncertainty, as reported in Table VIII. Figure 2 illustrates the relation-

TABLE VIII: Correlation between trust measures and predictive uncertainty.

Metric	Pearson	Spearman
CDT vs Uncertainty	-0.847	-0.333
CERA vs Uncertainty	-0.335	-0.294
CAS vs Uncertainty	-0.873	-0.404

ship between CDT and predictive uncertainty. Higher-trust predictions were generally associated with lower uncertainty. CDT showed a Pearson correlation of -0.847 , while CAS showed the strongest inverse association at -0.873 . Because uncertainty reliability is an explicit component of CDT and therefore also contributes to CAS, these findings should be interpreted as evidence of internal consistency of the trust construction rather than independent external validation.

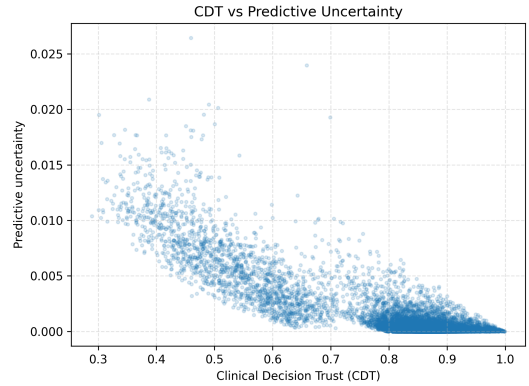


Fig. 2: Relationship between Clinical Decision Trust (CDT) and predictive uncertainty on the held-out pediatric test partition. Higher CDT values generally correspond to lower MC-dropout uncertainty.

C. Experiment 1: Selective Prediction Analysis

Figure 3 illustrates the coverage-reliability trade-off obtained when increasingly restrictive CDT thresholds are used to determine which predictions are accepted automatically. The corresponding operating points are reported numerically in Table IX.

At full coverage, Trust-Sleep accepted all 23,956 test epochs and achieved an accepted-case accuracy of 0.956, corresponding to an error rate of 4.40% and an apnea recall of 0.538. Increasing the acceptance threshold to $CDT \geq 0.60$ reduced coverage to 94.8% while increasing accepted-case accuracy to 0.973 and reducing the error rate to 2.69%. At $CDT \geq 0.70$, coverage was 92.7%, accepted-case accuracy increased further to 0.978, and the error rate decreased to 2.18%. At $CDT \geq 0.80$, 89.4% of the test epochs remained eligible for automatic acceptance, with an accuracy of 0.981 and an error rate of 1.95%.

A more restrictive operating point of $CDT \geq 0.85$ retained 60.9% of the test epochs and produced the highest accepted-case accuracy among the representative operating points in Table IX, 0.983, corresponding to an error rate of 1.71%. At $CDT \geq 0.90$, coverage fell sharply to 14.3%; accepted-case accuracy remained high at 0.975, although the error rate increased slightly to 2.52%. This non-monotonic behavior at very low coverage indicates that increasingly aggressive rejection does not necessarily produce continued improvements once only a small subset of cases remains.

These results show that prediction errors are disproportionately concentrated among lower-trust epochs. In particular, substantial improvements in accepted-case accuracy can be obtained while retaining approximately 89-95% coverage for CDT thresholds between 0.60 and 0.80. However, this improvement is accompanied by an important sensitivity trade-off. Accepted-case apnea recall decreased from 0.538 at full coverage to 0.283 at $CDT \geq 0.60$, 0.260 at $CDT \geq 0.70$, and 0.216 at $CDT \geq 0.80$. Therefore, CDT-based selective prediction should be interpreted as a mechanism for controlling automation and prioritizing lower-trust cases for clinician

review, rather than as a mechanism for safely excluding pediatric apnea events.

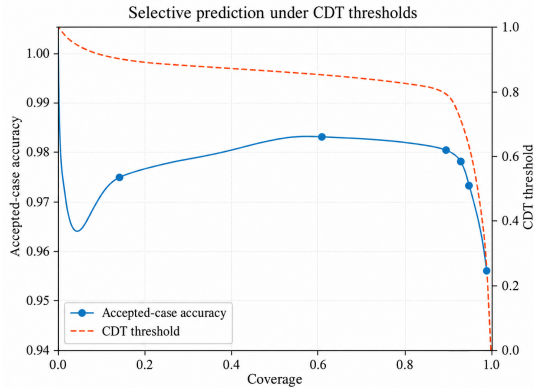


Fig. 3: Selective prediction under increasingly restrictive CDT thresholds. The figure visually summarizes the coverage–accuracy relationship. Exact operating-point values are reported in Table IX. Higher CDT thresholds generally increase accepted-case accuracy and reduce error, but also reduce coverage and accepted-case apnea recall.

TABLE IX: Selective-prediction operating points on the held-out pediatric test partition. Coverage denotes the fraction of test epochs retained for automatic acceptance at each CDT threshold.

CDT threshold	Coverage	Accuracy	Error rate	Apnea recall
0.00	100.0%	0.9560	4.40%	0.5379
0.60	94.8%	0.9731	2.69%	0.2834
0.70	92.7%	0.9782	2.18%	0.2605
0.80	89.4%	0.9805	1.95%	0.2159
0.85	60.9%	0.9829	1.71%	0.2284
0.90	14.3%	0.9748	2.52%	0.2522

D. Experiment 1: Error Stratification by CDT

Figure 4 addresses RQ3 by comparing misclassification rates across low-, medium-, and high-CDT groups. The low-CDT group ($CDT < 0.60$) contained 5.19% of test epochs and had an error rate of 35.7%. The medium-CDT group contained 5.43% of epochs and had an error rate of 14.8%. In contrast, the high-CDT group ($CDT \geq 0.80$) contained 89.4% of epochs and had an error rate of only 1.95%. Prediction errors were therefore strongly concentrated among lower-trust epochs.

E. Experiment 1: Explanation Reliability Analysis

CERA showed a modest positive association with correctness (Pearson 0.156; Spearman 0.154). Mean CERA was 0.733 among correct predictions and 0.709 among incorrect predictions. Among the automated CERA components, temporal consistency averaged 0.867 for correct predictions versus 0.784 for incorrect predictions, and physiological plausibility averaged 0.397 versus 0.322. By contrast, perturbation-stability alone was not higher for correct predictions in this run. These results indicate that the aggregate explanation-reliability score contains some decision-quality information,

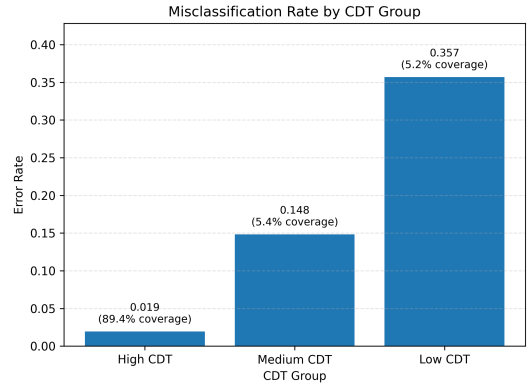


Fig. 4: Misclassification rate across low-, medium-, and high-CDT groups on the held-out test partition. Errors are concentrated within lower-trust samples.

but substantially less than CDT or CAS. The analysis remains automated and should not be interpreted as clinician-validated explanation quality.

F. Experiment 1: Privacy-Motivated Perturbation Robustness Analysis

Using the Laplace-perturbation protocol described in the Experimental Setup, Experiment 1 evaluated predictive behavior across ϵ values from 0.5 to 10. The corresponding numerical results are summarized in Table X. As expected,

TABLE X: Predictive robustness under epsilon-parameterized Laplace perturbations.

ϵ	F1	ROC-AUC	AP	ECE
0.5	0.546	0.906	0.571	0.0175
1.0	0.561	0.914	0.601	0.0124
2.0	0.584	0.919	0.619	0.0092
5.0	0.587	0.920	0.620	0.0069
10.0	0.593	0.920	0.620	0.0063

stronger perturbation was associated with lower predictive performance. The F1-score increased from 0.546 at $\epsilon = 0.5$ to 0.593 at $\epsilon = 10$, while ROC-AUC increased from 0.906 to approximately 0.920. Average precision and calibration similarly improved as perturbation magnitude decreased. These results characterize robustness to the evaluated perturbations only. They do not establish differential privacy, membership privacy, or any formal privacy guarantee.

G. Experiment 1: Comparison with Existing Methods

To place Trust-Sleep in context, the evaluation included matched logistic-regression and random-forest baselines using the standardized event descriptors, together with the Raw-Signal-only neural configuration. We additionally report PS-GMultiScaleCNN from the prior same-dataset study by Frenning [18]. The raw-signal branch used in Trust-Sleep follows the same multiscale design principle, with three parallel temporal convolutional pathways using kernel sizes 3, 7, and 15. The two models are therefore closely related architecturally,

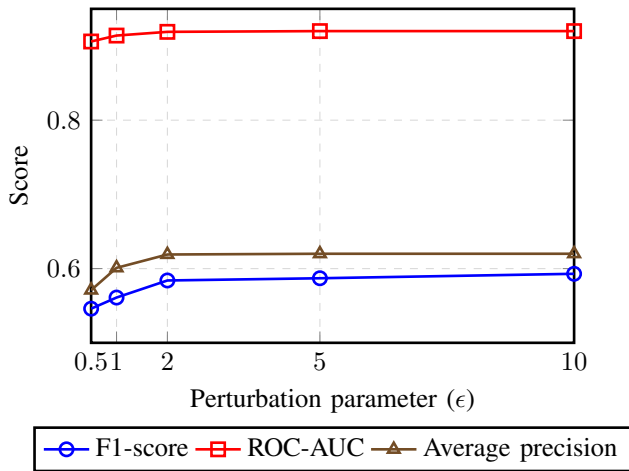


Fig. 5: Predictive behavior under epsilon-parameterized Laplace perturbations. The experiment measures perturbation robustness and is not a formal differential-privacy evaluation.

but their reported results were obtained under different experimental configurations.

In Frenning’s architecture-comparison experiment, PSG-MultiScaleCNN used the available CANOP PSG signals (chest, abdomen, nasal airflow, SpO₂, and pulse), per-recording normalization, weighted sampling, a fixed decision threshold of 0.5, and the training/validation/test procedure used in that study. The model achieved an accuracy of 0.950 and an F1-score of 0.450, with precision 0.32 and recall 0.74. In the present evaluation, Trust-Sleep achieved a similar overall accuracy of 0.956 but a substantially higher F1-score of 0.588, with precision 0.649 and recall 0.538. Thus, the main difference in predictive behavior is not a large change in overall accuracy, but a more balanced precision–recall trade-off on the imbalanced pediatric test set.

Several methodological differences may contribute to this performance difference. The present study uses four PSG channels, train-only preprocessing statistics, AdamW optimization, dedicated probability calibration and threshold-selection partitions, and a recording-disjoint held-out test set. In addition, Trust-Sleep augments the multiscale raw representation with a gated 52-dimensional event-centric representation. Because these changes were introduced together, the improvement over the previous PSGMultiScaleCNN result cannot be attributed to any single factor.

The contribution of the event-centric branch can, however, be assessed directly within the present study. Under the same split, preprocessing, optimization, calibration, and evaluation protocol, the Raw-Signal-only configuration achieved an F1-score of 0.521, whereas the Full Trust-Sleep model achieved 0.588 in Experiment 1. More importantly, the five-seed controlled comparison in Experiment 2 increased ensemble F1-score from 0.546 to 0.598 and ROC-AUC from 0.910 to 0.934 when the gated event-centric branch was added. These controlled results provide direct evidence that the event-centric representation contributes predictive information beyond the shared multiscale raw-signal backbone, while the remaining difference from Frenning’s earlier result may also

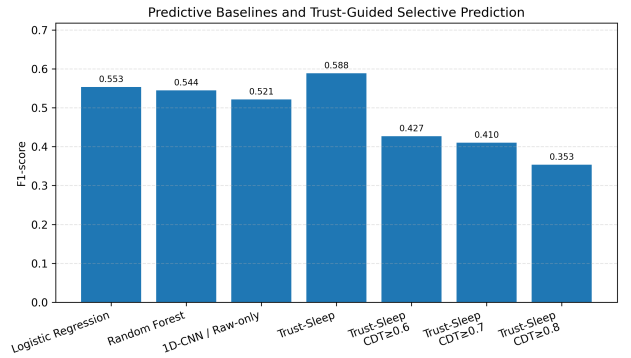


Fig. 6: Full-coverage baseline F1-scores together with selected CDT-threshold operating points for Trust-Sleep. Selective-prediction results involve reduced coverage and should be interpreted as operating points rather than direct full-coverage classifier comparisons.

reflect changes in preprocessing, optimization, calibration, and operating-threshold selection. Table XI summarizes the full-coverage results. Among the methods evaluated on the same

TABLE XI: Comparison with representative baseline methods. PSGMultiScaleCNN uses the same underlying model architecture and dataset source, but its reported result was obtained under a different experimental protocol and is therefore included as a contextual reference.

Method	Accuracy	F1	ROC-AUC
Logistic Regression	0.950	0.553	0.915
Random Forest	0.949	0.544	0.919
1D-CNN / Raw-only	0.950	0.521	0.901
PSGMultiScaleCNN [18]	0.950	0.450	–
Trust-Sleep	0.956	0.588	0.928

held-out test partition, Trust-Sleep achieved the highest F1-score and ROC-AUC. These comparisons provide predictive context rather than a fully controlled architectural ablation; the controlled contribution of the event-centric branch is examined separately in Experiment 2. The purpose of Trust-Sleep nevertheless remains broader than full-coverage classifier ranking: the framework additionally exposes uncertainty, conformal reliability, explanation-reliability quantities, and selective review behavior. Figure 6 places the conventional baselines beside the selective-prediction operating points. The latter should not be interpreted as directly superior classifiers, because selective filtering changes coverage and class composition.

1) Experiment 1: Capability Comparison: The predictive comparison in Table XI is complemented by the qualitative capability comparison in Table XII. The baseline models provide point predictions, whereas Trust-Sleep additionally reports uncertainty-related, conformal, explanation-reliability, selective-prediction, and trust-calibration outputs as defined earlier. This capability comparison is intended to clarify the broader scope of Experiment 1 rather than to imply that every trust-related component independently improves classifier accuracy.

TABLE XII: Capability comparison between Trust-Sleep and representative baseline approaches.

Method	Uncertainty Quantification	Conformal Prediction	Explanation Reliability	Selective Prediction	Trust Calibration
1D CNN	No	No	No	No	No
Logistic Regression	No	No	No	No	No
Random Forest	Partial	No	No	No	No
Trust-Sleep	Yes	Yes	Yes	Yes	Yes

H. Experiment 2: Five-Seed Ablation Studies

Following the framework-level analyses in Experiment 1, Experiment 2 examined whether the event-centric branch contributed predictive information beyond raw physiological waveforms. Full and Raw-Signal-only were trained using the same frozen recording splits, random seeds, preprocessing, optimization, early stopping, calibration procedures, and final test partition. The only intended predictive difference between the two configurations was the presence or absence of the gated event branch.

1) *Without Event Features*: Table XIII summarizes the held-out performance of the five-seed ensembles. Relative to Raw-Signal-only, the Full ensemble achieved higher ROC-AUC (0.934 vs. 0.910), average precision (0.645 vs. 0.584), F1-score (0.598 vs. 0.546), balanced accuracy (0.761 vs. 0.742), precision (0.674 vs. 0.596), recall (0.538 vs. 0.504), and specificity (0.984 vs. 0.979). The Full ensemble also achieved a lower Brier score and slightly lower ECE.

Table XIV reports the mean and standard deviation across the five independently trained seed-level models. Across seeds, the Full model remained more accurate and generally more stable. Its ROC-AUC standard deviation was 0.0022, compared with 0.0113 for Raw-Signal-only, and its average-precision standard deviation was 0.0116 compared with 0.0438. F1 variability was similar but still modest for both models.

The paired recording-bootstrap results are shown in Table XV. The intervals excluded zero for accuracy, balanced accuracy, precision, specificity, F1-score, ROC-AUC, average precision, and the Brier-score improvement. The recall difference was positive (0.0336), but its 95% interval included zero, so a robust recall improvement is not claimed. Table XVI provides the ensemble confusion matrices. Compared with Raw-Signal-only, the Full ensemble identified 47 additional positive epochs while producing 115 fewer false positives. The improvement was therefore not due simply to a more liberal positive-prediction tendency. Taken together, these findings indicate that the gated event representation contributed complementary physiological information and improved predictive performance across random initialization.

2) *Without Uncertainty Estimation*: Uncertainty estimation was not implemented as a separately trained predictive model. Rather, it operated after classification and supplied reliability information to the trust and selective-decision layers. Removing this component would therefore leave the underlying calibrated classifier unchanged while eliminating the mechanism used to quantify prediction instability across stochastic evaluations. Consequently, no numerical change in ROC-AUC, average precision, or F1-score is attributed specifically to uncertainty estimation within the Full-versus-Raw predictive ablation.

3) *Without Conformal Prediction*: Conformal prediction was likewise a post-prediction reliability component rather than a separately trained classifier. Removing it would not alter conventional discrimination metrics, but would eliminate set-valued predictions, empirical coverage monitoring, and the conformal contribution to the trust layer. Table XVII summarizes the held-out conformal results. At nominal $\alpha = 0.10$, the Full ensemble achieved empirical epoch-level coverage of 0.9082 and Raw-Signal-only achieved 0.9077, both close to and slightly above the nominal 0.90 target on this held-out partition.

Neither model produced two-class ambiguous sets; all non-singleton cases were empty sets. These cases should be interpreted as abstentions or escalation signals rather than converted into forced class decisions. The observed coverage supports the calibration behavior on this held-out same-cohort partition, but it should not be generalized as a distribution-free guarantee under future site or population shift.

4) *Without Clinical Explanation Reliability Assessment (CERA)*: CERA evaluates explanation stability, temporal consistency, reliability consistency, physiological plausibility, and the automated interpretability heuristic used in Experiment 1. A controlled numerical CERA-removal experiment was not included in the reported predictive ablation. Accordingly, no improvement in ROC-AUC, F1-score, or calibration is attributed specifically to CERA. The present evidence for CERA comes from its association with correctness and from component-level behavior within Experiment 1.

5) *Without Clinical Decision Trust (CDT)*: CDT combines predictive confidence, uncertainty reliability, conformal reliability, physiological evidence, and signal quality into a unified decision-support quantity. Removing CDT would leave the calibrated classifier unchanged but remove the prespecified mechanism for trust-based acceptance, abstention, and clinician escalation. The held-out selective-prediction results show that higher CDT thresholds substantially reduce accepted-case error while also reducing apnea recall; this trade-off is the relevant evidence for the decision-support contribution of CDT rather than a change in ROC-AUC or average precision.

6) *Synthesis of Key Findings*: Taken together, the two experiments suggest that Trust-Sleep is best interpreted as a trust-aware decision-support framework rather than a standalone diagnostic system. Experiment 2 showed that adding the gated event-centric branch improved predictive performance relative to Raw-Signal-only across F1-score, ROC-AUC, average precision, precision, balanced accuracy, specificity, Brier score, and overall accuracy, with paired recording-level intervals excluding zero for these effects. Experiment 1 further showed that higher CDT and CAS values were associated with greater prediction correctness and lower predictive uncertainty, and that selective prediction could substantially reduce accepted-case error by referring lower-trust cases for review. The same selective policy reduced accepted-case recall, however, emphasizing that abstention should support clinician prioritization rather than autonomous exclusion of pediatric apnea. Conformal coverage on the held-out same-cohort partition was close to the nominal target for both ensembles. These findings support Trust-Sleep as a clinician-facing framework

TABLE XIII: Held-out performance of the five-seed ensembles. Thresholds were selected only on the dedicated threshold-selection partition.

Model	Acc.	Bal. Acc.	Prec.	Recall	Spec.	F1	ROC-AUC	AP	Brier ↓	ECE ↓
Full	0.958	0.761	0.674	0.538	0.984	0.598	0.934	0.645	0.0324	0.0077
Raw-Signal-only	0.951	0.742	0.596	0.504	0.979	0.546	0.910	0.584	0.0357	0.0083

TABLE XIV: Held-out seed-level performance, reported as mean $\pm$ standard deviation over five independent runs.

Model	Accuracy	Bal. Acc.	Precision	Recall	Specificity	F1	ROC-AUC	AP
Full	0.957 $\pm$ 0.002	0.746 $\pm$ 0.024	0.687 $\pm$ 0.055	0.507 $\pm$ 0.052	0.985 $\pm$ 0.005	0.580 $\pm$ 0.020	0.930 $\pm$ 0.002	0.630 $\pm$ 0.012
Raw-Signal-only	0.950 $\pm$ 0.006	0.714 $\pm$ 0.016	0.617 $\pm$ 0.092	0.447 $\pm$ 0.040	0.981 $\pm$ 0.009	0.513 $\pm$ 0.018	0.901 $\pm$ 0.011	0.549 $\pm$ 0.044

TABLE XV: Paired recording-bootstrap differences between Full and Raw-Signal-only. Positive values favor Full except for Brier score, where negative values favor Full.

Metric	Difference	95% CI
Accuracy	0.0068	[0.0045, 0.0093]
Balanced accuracy	0.0193	[0.0010, 0.0429]
Precision	0.0783	[0.0454, 0.1081]
Recall	0.0336	[-0.0061, 0.0816]
Specificity	0.0051	[0.0019, 0.0087]
F1-score	0.0521	[0.0304, 0.0866]
ROC-AUC	0.0246	[0.0146, 0.0340]
Average precision	0.0609	[0.0413, 0.0871]
Brier score	-0.00334	[-0.00564, -0.00149]

TABLE XVI: Held-out ensemble confusion matrices.

Model	TN	FP	FN	TP
Full	22,192	364	647	753
Raw-Signal-only	22,077	479	694	706

TABLE XVII: Held-out conformal-prediction results.

Model	$\hat{q}$	Coverage	Singleton	Empty	Doubleton
Full	0.1896	0.9082	0.9270	0.0730	0.0000
Raw-Signal-only	0.1823	0.9077	0.9303	0.0697	0.0000

for prioritization, uncertainty communication, and selective review, while prospective and independent-site validation remain necessary before clinical deployment.

VI. DISCUSSION

A. Clinical Interpretation

The present findings suggest that Trust-Sleep is best interpreted as a clinician-facing screening and prioritization framework rather than as a standalone diagnostic system. Across the five-seed ablation in Experiment 2, the Full configuration consistently outperformed Raw-Signal-only, indicating that event-centric physiological summaries contributed information beyond the raw waveforms alone. At the same time, recall remained moderate, which implies that a substantial fraction of apnea epochs could still be missed if the model were used without clinical oversight. In practical terms, the framework appears better suited to highlighting potentially relevant epochs and supporting review efficiency than to excluding apnea or establishing patient-level diagnosis on its own.

B. Human-AI Collaboration

A central motivation of Trust-Sleep is to support selective, human-supervised use of model outputs. In Experiment 1, higher CDT and CAS values were associated with greater

prediction correctness and lower predictive uncertainty, and lower-trust cases contained a larger share of prediction errors. These patterns support the use of trust scores as operational criteria for deciding when automated output may be useful and when clinician review should be prioritized. The trust-policy analyses further suggest that the practical value of the framework lies less in replacing clinical judgment than in structuring escalation decisions, identifying lower-confidence cases, and making uncertainty more explicit within the workflow.

C. Conformal Reliability and Abstention

The conformal analyses revealed both utility and limitation. Empirical coverage on the held-out recording-disjoint test partition was 0.9082 for the Full ensemble and 0.9077 for Raw-Signal-only at nominal 0.90, indicating close agreement with the target within this same-cohort evaluation. At the same time, non-singleton outputs were empty sets rather than two-class ambiguous sets, which means that these cases should be treated as abstentions or escalation signals rather than converted into forced class predictions. Accordingly, the conformal component is useful for flagging potentially unreliable cases, but the present results do not establish unconditional coverage under future distribution, site, or population shift.

VII. LIMITATIONS

This study has several important limitations. First, all reported analyses were derived from a single pediatric source cohort and therefore do not establish external multi-center generalization. The source LDS contains children aged 1–12 years with underlying disorders and no healthy control group, so performance may differ in community screening populations, other age distributions, or other clinical sites. Second, although the recording-level split prevents recording overlap among training, calibration, and test subsets, the final test partition contains 19 recording identifiers; larger independent pediatric cohorts are needed to reduce uncertainty about population-level generalization. Third, recall remained moderate even for the Full ensemble, so the model may miss clinically relevant apnea epochs if used without human oversight. Fourth, the explanation-reliability component was assessed through automated criteria such as perturbation stability, temporal consistency, reliability consistency, physiological plausibility, and an automated interpretability heuristic; it was not validated through clinician-rated interpretability and should not be interpreted as such. Fifth, the epsilon-parameterized Laplace experiment evaluates perturbation robustness only and does

not establish a formal differential-privacy guarantee. Finally, epoch-level screening metrics do not directly establish patient-level diagnostic accuracy or clinical utility. Prospective validation with age-aware pediatric analysis, patient-level endpoints, and clinician review is therefore required before deployment.

VIII. ETHICAL, DATA-GOVERNANCE, AND CLINICAL CONSIDERATIONS

The pediatric sleep-study data used in this work were collected under the applicable institutional and regulatory procedures for clinical research. The study protocol and use of the data were approved by the relevant ethics. Where required, appropriate consent or other legal basis for the collection and secondary research use of the data was obtained in accordance with the governing study protocol.

All data used for the present analysis were de-identified before model development and evaluation. Access to the underlying clinical recordings is restricted to authorized researchers and is subject to the data-governance requirements of the originating study. Because the dataset contains sensitive pediatric clinical information, the raw recordings cannot be made publicly available. Access may be considered only through the applicable institutional data-access procedures and subject to relevant ethical, legal, and data-protection requirements.

Trust-Sleep is intended as a clinical decision-support framework rather than an autonomous diagnostic system. The reported trust scores should not be interpreted as guarantees of correctness, and the explanation-related quantities have not been clinically validated as measures of human interpretability. Responsible deployment would require prospective validation, site-specific assessment, appropriate clinician oversight, and continued monitoring under the target clinical workflow.

IX. REPRODUCIBILITY AND TRANSPARENCY

Several design choices were introduced to improve reproducibility and auditability. The analysis treats the pediatric export as a single master dataset and uses one fixed recording-level partition shared by Experiment 1 and Experiment 2: 60% training, 10% early-stopping validation, 5% Platt calibration, 5% threshold selection, 10% conformal calibration, and 10% final testing. Preprocessing parameters were estimated only on the training partition. The final test partition was not used for model fitting, model selection, probability calibration, threshold selection, or conformal calibration. Split assignments, preprocessing statistics, checkpoints, calibration artifacts, prediction files, and recording-level bootstrap outputs were retained as an audit trail. The reported runs were executed in Google Colab using a CUDA-enabled NVIDIA Tesla T4 GPU. The five prespecified random seeds were {271828, 314159, 161803, 141421, 173205}. The recorded environment used Python 3.13.15, NumPy 2.1.3, pandas 2.2.3, and PyTorch 2.11.0 with CUDA 13.0. Subject to the applicable data-governance restrictions, the model architecture, evaluation code, configuration files, split metadata, and non-sensitive aggregate experimental outputs may be shared to support methodological reproducibility.

X. CONCLUSION

This work presented Trust-Sleep, a trust-aware clinical decision-support framework for pediatric sleep apnea screening that integrates physiological modeling, a gated event-centric representation, trust scoring, conformal reliability, and selective prediction. Experiment 2 showed that the event-enhanced full ensemble improved the F1 score, ROC-AUC, average precision, precision, balanced accuracy, specificity, Brier score, and overall accuracy relative to an otherwise matched Raw-Signal-only ensemble, while the recall improvement was not statistically conclusive. Experiment 1 showed that higher CDT and CAS values were associated with greater correctness and lower uncertainty and that selective prediction reduced accepted-case error by referring lower-trust cases for review, although higher trust thresholds also reduced accepted-case apnea recall. Conformal coverage on the held-out test partition was close to the nominal 90% target. Trust-Sleep should therefore be viewed as a clinician-facing framework for uncertainty communication, prioritization, and selective review rather than as a standalone diagnostic system. Future work should validate the framework on independent pediatric cohorts, assess performance across age and disease subgroups, and evaluate trust and explanation reliability prospectively within clinical workflows.

BROADER IMPACT

Trust-Sleep aims to support safer and more transparent AI-assisted sleep apnea screening by combining prediction with uncertainty, reliability assessment, and selective referral for clinician review. Such a framework may help prioritize difficult cases, make uncertainty more explicit, and reduce unnecessary manual review of higher-confidence cases. However, trust scores and explanation-reliability measures should not be interpreted as guarantees of correctness or clinical validity, and selective prediction introduces a trade-off between reliability and coverage that may result in clinically relevant apnea events being deferred or missed. Performance may also vary across age groups, disease severity, demographic subgroups, recording conditions, and clinical sites, creating risks of unequal error, abstention, or referral rates; subgroup-level fairness should therefore be explicitly evaluated rather than inferred from aggregate performance. Accessibility is also important, since the practical benefit of trust-aware decision support may depend on computational resources, availability and quality of PSG signals, clinician expertise, and whether the interface communicates uncertainty in a clear and usable manner across different clinical settings. Responsible use therefore requires appropriate data governance, external and subgroup-level validation, fairness and calibration monitoring, site-specific assessment, attention to accessibility and workflow integration, and continued clinician oversight. Trust-Sleep is intended to support human-AI collaboration and clinical prioritization rather than replace clinical judgment or serve as an autonomous diagnostic system.

USE OF GENERATIVE ARTIFICIAL INTELLIGENCE

Generative artificial intelligence tools, including OpenAI ChatGPT, were used during manuscript preparation to assist

with language editing, improving clarity and organization, LaTeX formatting, and selected programming and documentation tasks. Such tools were used as assistive tools rather than as autonomous scientific contributors. All methodological choices, experimental designs, analyses, numerical results, interpretations, figures, tables, references, and conclusions were reviewed and verified by the authors. AI-generated suggestions were edited or rejected where necessary, and the authors take full responsibility for the accuracy, originality, and integrity of the final manuscript. Generative AI systems were not treated as authors and did not make independent decisions regarding the scientific conclusions of the study.

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